



Velocity Nexus Prime: Comprehensive Vehicle Management System

Velocity Nexus Prime is a professional-grade vehicle management system designed to streamline modern dealership operations. Developed by Muhammad Danish Anwar & Abdul Manan Arif at Lahore Garrison University, this project showcases an end-to-end solution for inventory, sales, customer relationships, and business analytics.

Key Features: Driving Dealership Efficiency



Inventory Management

Real-time tracking of vehicles across multiple dealerships, ensuring accurate stock levels and availability.



Sales Processing

Automated workflows for tax, commission, and final sales calculations, reducing errors and saving time.



Customer CRM

Comprehensive management of customer profiles, purchase history, and lifetime value for personalized engagement.



Business Analytics

Real-time dashboards providing insights into sales trends, performance metrics, and operational efficiency.

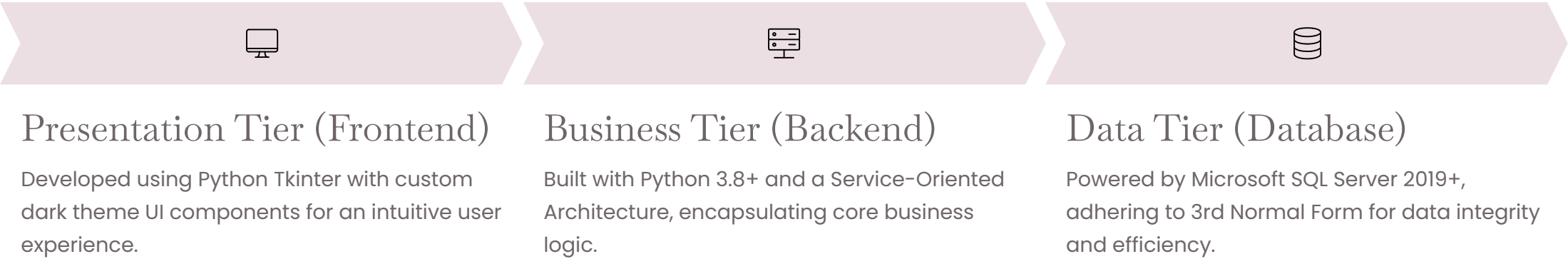


Robust Security

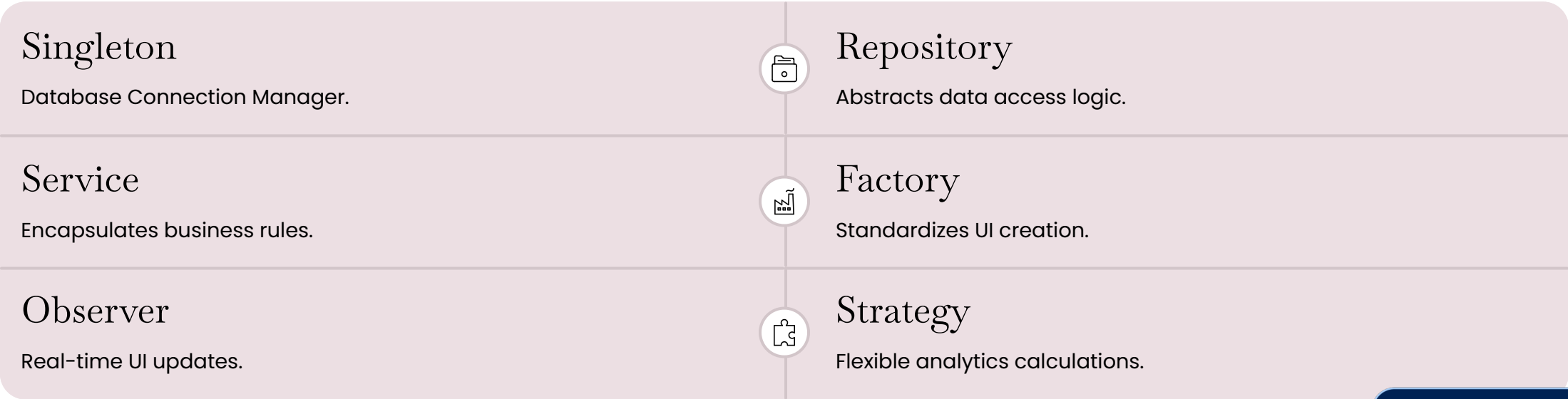
Role-based access control with comprehensive audit logging, protecting sensitive data and ensuring compliance.

System Architecture: A Foundation of Modularity

The Velocity Nexus Prime system follows a strict 3-Tier Architecture to ensure modularity, scalability, and maintainability, separating concerns for optimal performance.



Design Patterns Implemented



Database Architecture & Advanced SQL Logic

The database is the backbone of Velocity Nexus Prime, designed with meticulous adherence to BCNF principles and robust SQL implementations for data integrity and complex business logic.

Core Stored Procedures

- sp_ProcessCompleteSale

Handles entire sales transactions with ACID properties, including tax calculation (17% standard, 35% imported), commission, status updates, customer statistics, and audit logging.

- sp_GenerateFinancialReport

Generates a comprehensive 3-part financial report (Summary, Payment Methods, Daily Trends) using UNION ALL.

SQL Functions

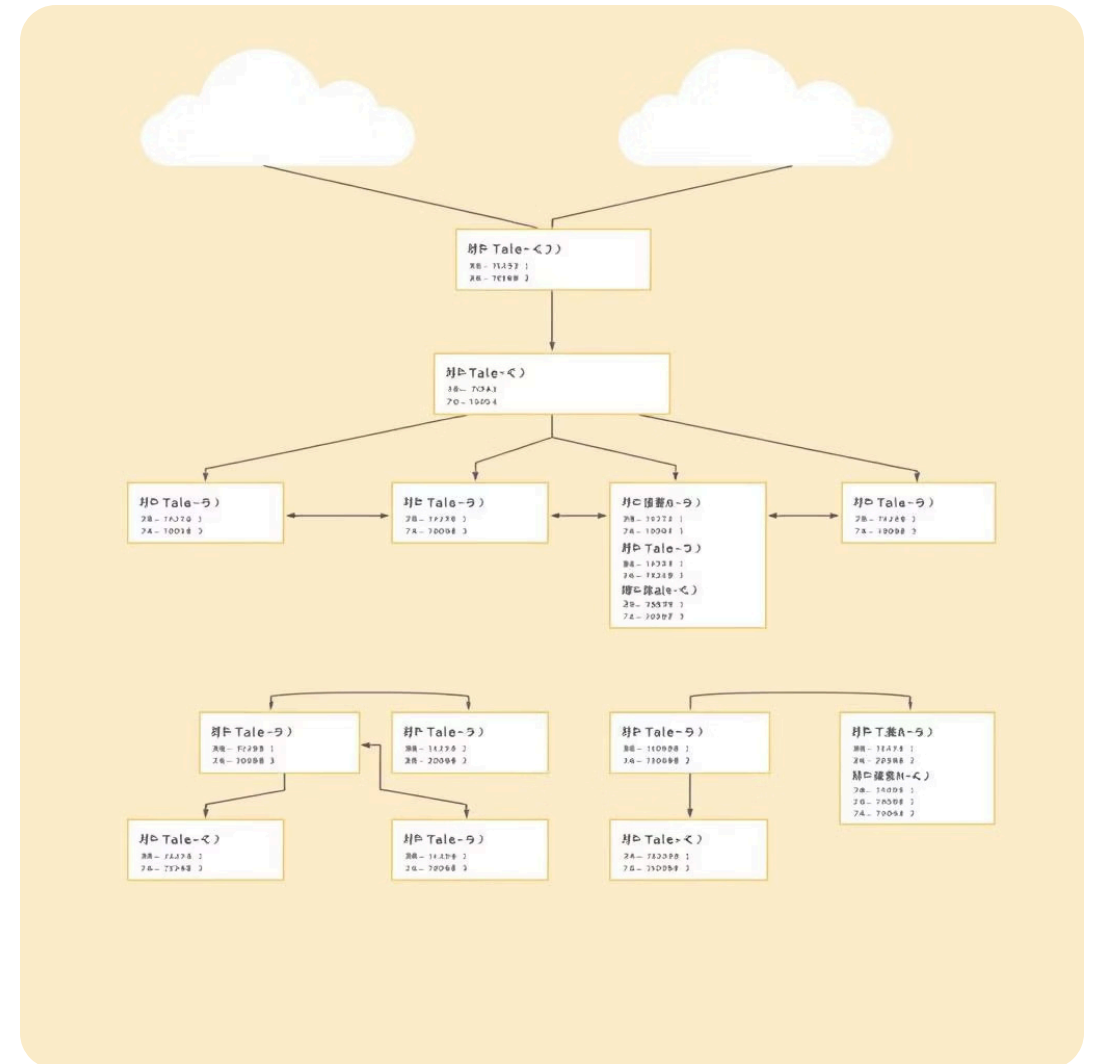
- fn_CalculateDepreciation

Calculates current vehicle value based on age (5% per year) and mileage factors.

- fn_CalculateCustomerLTV

Computes Customer Lifetime Value from past sales and projected future interactions.

Normalization Process (BCNF)



01

1NF

Eliminated repeating groups.

02

 $2NF$

Removed partial dependencies.

03

3NF

Removed transitive dependencies.

04

BCNF

Ensured every determinant is a candidate key.

Software Construction & Future Outlook

Adhering to crucial software development principles and employing rigorous testing, Velocity Nexus Prime is built for reliability and continuous improvement.

Layman's Laws of Software Development

Changing Requirements

Anticipating and adapting to evolving project needs.

Hidden Complexity

Simplifying features without compromising underlying intricacy.

Debugging Time

Acknowledging and planning for the time investment in bug resolution.

Documentation Debt

Prioritizing clear and comprehensive documentation.

Deployment & Academic Achievements



- **Deployment Pipeline:** One-click LAUNCHER.bat script for easy setup.
- **Configuration Management:** Utilizes python-dotenv for environment variables.
- **CI/CD:** GitHub Actions configured for automated testing on every push.
- **Functional Completeness:** 100% of requirements implemented (Sales, Inventory, CRM).
- **Performance:** Sub-second response times for queries on over 1000 records.
- **Quality:** Integrated 3-Tier Architecture with 6 robust Design Patterns.